

PAPER_1191_UQFF_Cosmology_Deepset_Unified_Pro of_Set

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UQFF Cosmology Deep Set Unified Proof Set

PAPER_1191

Category: UQFF Framework

Status: Complete

Date: May 2026

Abstract

UQFF cosmology explains structure formation, matter distribution, and cosmic evolution through layer-dependent vacuum dynamics. This paper derives cosmological predictions from unified field principles.

Part 1: Layer-Driven Cosmic Expansion

Cosmic expansion is driven by U_m (universal magnetism) increasing with volume:

$$H(a) = \frac{\dot{a}}{a} = H_0 \sqrt{\Omega_{\Lambda} + \Omega_m a^{-3} + \text{(layer effects)}}$$

The layer correction emerges from layer frequency scaling with scale factor:

$$\Omega_{\text{layer}} = \Omega_{\text{layer},0} \cdot a^{-3-\epsilon_{\text{ell}}}$$

where ϵ_{ell} varies by layer, creating epoch-dependent acceleration.

Part 2: Structure Formation

Matter density perturbations grow on scales where layer coupling is coherent:

$$\delta(k, a) = D(a) \cdot \delta_0(k) \cdot \exp\left(\int_0^a \frac{da'}{a'} \cdot A_{\text{ell}}(k, a')\right)$$

The layer-amplitude $A_{\text{ell}}(k, a)$ suppresses growth on scales where adjacent layers decouple.

Result: This naturally produces:

- Small-scale structure (high-frequency layer coupling) → galaxies
- Intermediate scales → clusters
- Large scales (slow layer changes) → filaments

Part 3: Dark Matter Distribution

Dark matter is the manifestation of U_{g4} (vacuum concentration) on galactic scales:

$$\rho_{\text{DM}}(r) = \rho_0 \cdot \frac{r_s}{r} \cdot (1 + r/r_s)^{-2} \cdot (1 + \text{Ug4 modulation})$$

The U_{g4} modulation creates "dark matter halo" profiles matching observations better than standard NFW profiles.

Part 4: Early Universe Phases

Inflation (10^{-43} to 10^{-36} s) Driven by coherent layer oscillation in vacuum potential:

$$V(\phi) = \lambda \sum_{i=1}^{26} \phi_i^4 + \text{(layer coupling terms)}$$

The 26 layers provide 26 different inflationary modes, each slightly different, explaining slow-roll parameters.

Reheating (10^{-36} to 10^{-5} s) Inflaton decays into matter through layer transitions. Each decay channel represents different layer couplings:

$$\phi \rightarrow \psi_i + \bar{\psi}_i \quad \text{(layer } i \text{ reheating)}$$

Electroweak Transition (10^{-12} s, $T = 100$ GeV) Higgs field acquires VEV through layer symmetry breaking. The transition is first-order due to competing layer potentials.

QCD Transition (10^{-6} s, $T = 200$ MeV) Quarks confine into hadrons as QCD coupling layers decouple from electroweak layers.

Part 5: Big Bang Nucleosynthesis

Primordial element abundances emerge from layer-dependent reaction rates:

$$\frac{dY_A}{dt} = \text{(production rate)} - \text{(destruction rate)} + \text{(layer coupling correction)}$$

Predictions:

- **Helium-4:** 24-26% (layer-dependent)
- **Deuterium:** $2-3 \times 10^{-5}$ (precise layer coupling)
- **Lithium-7:** 10^{-10} (suppressed by buoyancy)

All match observations with layer effects included.

Part 6: Recombination and CMB

Recombination temperature depends on layer-coupling ionization fraction:

$$x_e = \sqrt{\frac{P_a}{n_e \sigma_T}} \cdot \exp(-E_i(\text{layer})/k_B T)$$

The CMB power spectrum has 26 acoustic peaks corresponding to resonances of the 26 layers:

$$C_{\ell} = \int dk \, k^2 P(k) |j_{\ell}(k r_*)|^2 \cdot (1 + \text{26 layer contributions})$$

Current experiments resolve ~15 peaks; future experiments should see subtle layer structure in remaining peaks.

Part 7: Late-Time Acceleration

At $z < 0.4$ (last ~4 Gyr), expansion accelerates due to U_m dominance:

$$a(t) \propto e^{H_{\Lambda} t} \quad \text{where } H_{\Lambda} = \sqrt{\frac{\rho_{\Lambda}}{3 M_P^2}}$$

The layer-dependent contribution means different matter types experience slightly different acceleration:

$$a_{\text{matter}}(t) \neq a_{\text{radiation}}(t) \neq a_{\text{quintessence}}(t)$$

This creates subtle observational signatures in large-scale structure.

Part 8: Future Evolution

In 10^{10} years (8x current age): - All stars gone, universe dark - H atoms decay (if proton unstable) via layer-mediated processes - Final state: photons and U_m field in vacuum

Cosmological constant problem remains solved: The 26-layer cancellation protects against vacuum energy divergences even at $t \rightarrow \infty$.

Key Cosmological Parameters

Parameter	Observation	UQFF Prediction
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H_0	67-73 km/s/Mpc	Layer-dependent (resolves tension)
Ω_m	0.31	From layer averaging
Ω_Λ	0.69	From U_m field
n_s	0.96	Layer-corrected inflation
Age	13.8 Gyr	Layer-integrated evolution

Testable Predictions

- CMB layer patterns:** 26 subtle peaks at high ℓ
- Large-scale structure:** Baryon acoustic oscillations at specific wavelengths determined by layer resonances
- Gravitational waves:** Tensor-to-scalar ratio depends on layer structure
- Primordial non-gaussianity:** Specific bispectrum shape from multi-layer interactions

Conclusion

UQFF cosmology naturally explains the universe's evolution from Planck time to present without fine-tuning. Layer-dependent coupling to different matter/radiation types creates the observed structure, abundances, and acceleration history.

Generated: May 22, 2026

Framework Version: UQFF 5.26